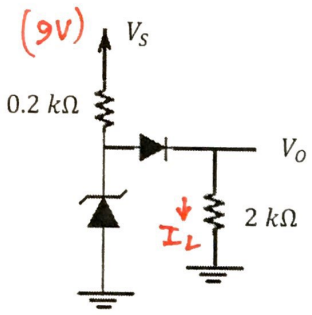
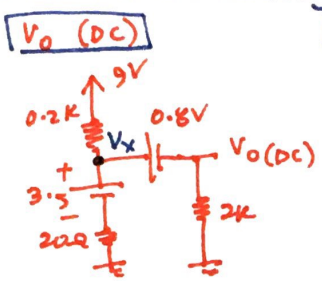
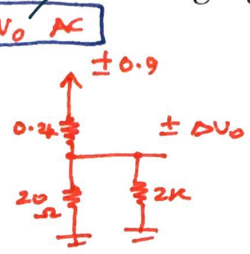
1883 — 75 minutes (40+30+30 points)

Name:	ID:	V_S = 9 V	V_{DD} = 9 V	Mark: /100

1) The circuit below uses a Zener diode with $V_{ZO} = 3.5\,V$, $r_z = 20\,\Omega$ and $I_{z_min} = 1.0\,mA$. The regular diode is modeled with $V_{DO} = 0.8\,V$ and $r_D = 0\,\Omega$ when forward-biased. (a) If the source V_S has a $\pm 10\%$ variation, calculate the percentage variation in the output voltage V_O. (b) Find the minimum load resistance that can be connected without losing regulation. Show all steps.







(a)	(b)

$V_{O(DC)} = V_X - 0.8$
$= 3.17\,V$

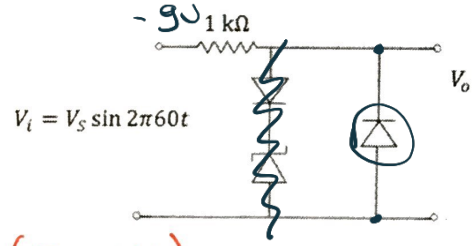
$\pm\Delta V_O = \pm(0.9)\dfrac{2k//20}{0.2\,k + 2k//20}$
$= \pm\,0.081\,V$

$\% \dfrac{\Delta V_O}{V_{O(DC)}} = 2.56\,\%$

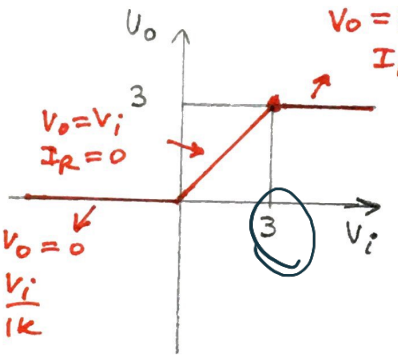
$I_L = \dfrac{0.9(9) - V_{O\,DC}}{0.2k} - 1$

$R_{L\,min} = V_O / I_L$

2) For the diode circuit below (a) Draw the transfer function (V_O vs. V_i) of the circuit. Show all labels on the curve. (b) Find the maximum and minimum current through the resistor. Use the ideal diode model. The Zener diode has $V_{ZO} = 3\,V$ and $r_z = 0\,\Omega$. Show all steps and calculations.



$(V_S = 9\,V)$



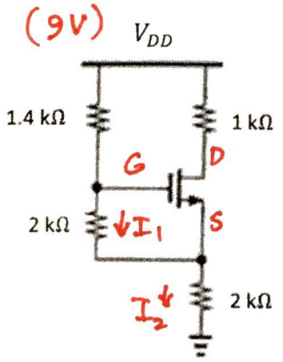
$I_R = \dfrac{V_i}{1k}$

I_{R_max}	I_{R_min}

$I_{R\,min} = -9\,mA$

$I_{R\,max} = 6\,mA$

3) The MOSFET in the circuit below has $V_t = 1$ V, k=1 mA/V^2 and λ=0. Given I_D=1.5 mA, analyze the circuit. Show all steps and calculations.



Assume saturation

$1.5m = 1m\,(V_{GS} - 1)^2$

$\Rightarrow V_{GS} = 2.22\,V$

$I_1 = V_{GS}/2k = 1.11\,mA$

$I_2 = I_1 + I_D = 2.61\,mA$

$V_S = I_2\,(2k) = 5.22\,V$

$V_G = 2.22 + 5.22 = 7.44\,V$

$V_D = 9 - 1.5m(1k) = 7.5\,V$

} saturation ✓

V_D	V_G	V_S

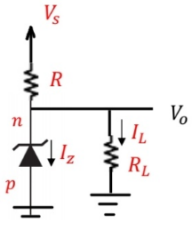
$\dfrac{9 - V_X}{0.2k} = \dfrac{V_X - 3.5}{20} + \dfrac{V_O}{2k}$

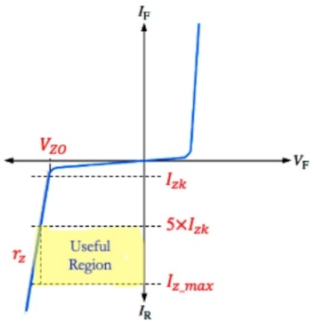
$V_Z = 6.8\ V$ *at* $5\ mA$, $r_z = 20\ \Omega$ and $I_{zk} = 0.2\ mA$.

$$Id = k \cdot (V_{GS} - V_t)^2$$

$$Id = k \cdot \left(2(V_{GS} - V_t) \cdot V_{DS} - V_{DS}^2\right)$$

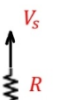
The resistance R should not be too high. Otherwise the diode will not get the minimum current for useful region?





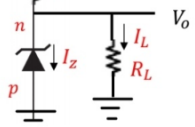
$$R_{max} = \frac{V_{s_min} - V_{ZO} - r_z\, I_{Z_min}}{I_{Z_min} + I_{L_max}}$$

4) Design a shunt regulator to get an output voltage around 10 V. Use the 1N7470. The input source is expected to have 20±25% V and the load can take current up to 20 mA.



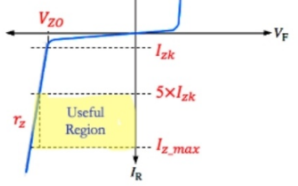
1N 7470: $V_Z = 10\ V$ *at* $25\ mA$, $r_z = 7\ \Omega$ and $I_{zk} = 1\ mA$





$$V_{ZO} = V_Z - I_Z\, r_z$$

$$V_{ZO} = 10 - (25\ m)(7) = 9.825\ V$$



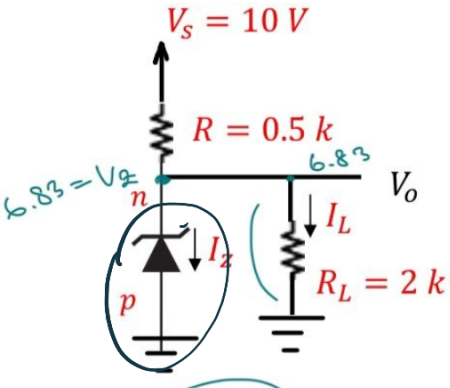
$$R_{max} = \frac{V_{s_min} - V_{ZO} - r_z\, I_{Z_min}}{I_{Z_min} + I_{L_max}}$$

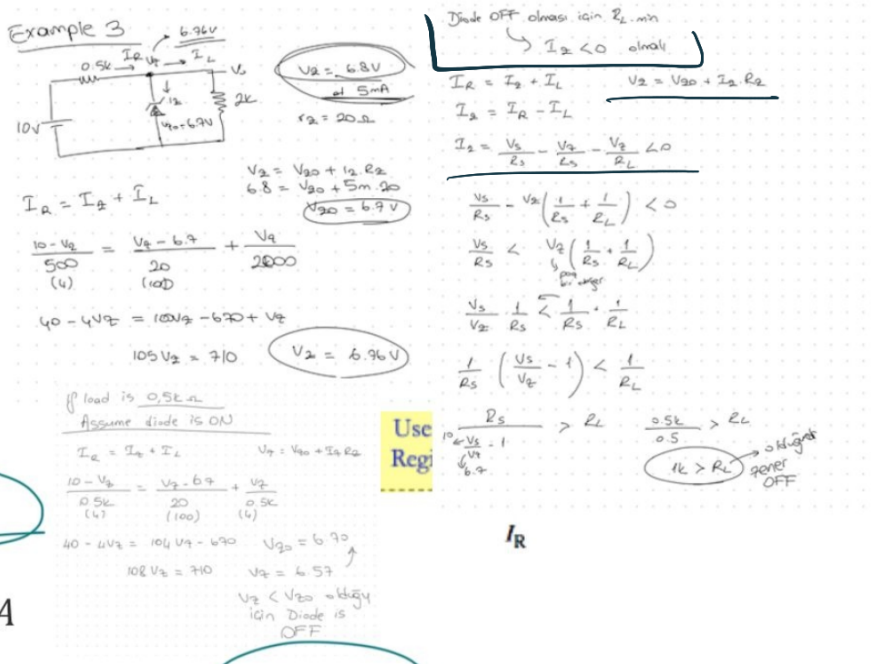
$$R_{max} = \frac{15 - 9.825 - (7)\,(5\ m)}{5\ m + 20\ m} = 205\ \Omega$$

$$R \le 205\ \Omega$$

---

3) The source now is fixed at 10 V, but a load of 2 kΩ is connected. What is the change in output voltage?





$$I_L = \frac{V_o}{R_L} = \frac{6.83}{2\ k} = 3.42\ mA$$

$$I_Z = 6.35 - 3.42 = 2.93\ mA$$

$$V_o = V_{ZO} + I_Z\, r_z = 6.7 + (2.93\ m)(20) = 6.76\ V$$

1% change in output

What if load is 0.5 k Ω?

$$I_L = 6.83/2k = 13.66\ mA$$

0.5k

Diode is OFF

What is R_{L_min}?